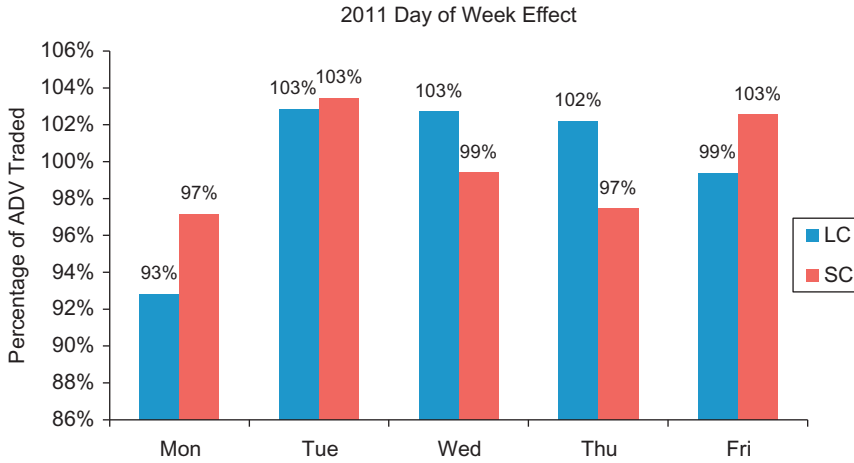




■ Figure 7.4 Day of Week Effect.

Step 3. Estimate the autoregressive parameter $\hat{\beta}$.

The autoregressive parameter is used to correct for persistence of volume over consecutive days. We found above average volume days were more likely to be followed by above average volume days and below average volume days were more likely to be followed by below average volume days. But the relationship was much more significant for above average volume days than for the below average volume days. This process is as follows:

- Estimate volume for the day based on the 10-day median plus the day of week adjustment.
- Compute the forecast error term as the difference between the actual volume on the day and the estimated volume. This difference is:
 - $\varepsilon(t) = V(t) - \text{Median}(t) \cdot \text{Day Of Week}(t)$
- Run a regression of the error term on its one-day lagged term, that is,
 - $\varepsilon(t) = \alpha + \beta \cdot \varepsilon(t - 1)$
- Compute the slope term β for large and small cap stocks.

Large cap stocks had a much larger degree of autocorrelation than small cap stocks. The average correlation of errors was $\beta_{\text{large}} = 0.452$ for large cap stocks and $\beta_{\text{small}} = 0.284$ for small cap stocks. After correcting for autocorrelation there was still a very slight amount of autocorrelation present, but this will have a negligible effect on our forecasts due to the constant term in our regression model.